

GREEN COMPUTING

Computer designs have improved markedly but not much has been done in terms of energy-efficiency and minimisation of power consumption of e-equipment



Seagate Barracuda Green hard drive

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'Go green' is today's slogan and tomorrow's mantra, and technology is not barred from it. Green technology has emerged as an important issue of policy and priority of the future society. The main goal is to produce and utilise technology that does not damage the environment and deplete earth's natural resources.

Green technology is thus an environment-friendly technology that envisages how to reduce waste, and how to produce innovative technologies towards new and renewable energy-generation and study of advanced materials. Due to ever increasing use of information technology (IT) in the society, green computing or green IT has become a crucial component of green technology.

The idea of green computing origi-

nated in 1992 following the 'energy star' programme. Energy star is an international standard designed and developed by a joint program of the US Environmental Protection Agency and the US Department of Energy. The standard was made in early 1990s to help customers save money and to protect the environment through energy-efficient products and practices. Although computers and monitors were the first labeled products, the energy star label is now applied to all major equipment, appliances and products. Green computing promises to offer eco-friendly computing with better and efficient use of IT resources, minimum use of unsafe materials and maximisation of energy conservation with reduced waste.

Reducing electronic waste or e-waste is one of the key challenges in realising eco- and environment-friendly technology. As per a report, "The United Nation Environment Programme (UNEP) estimated that worldwide, between 20 and 50 million tonnes of electrical and electronic goods which had come to the end of their lives were being thrown away every year. The latest UNEP report now estimates the annual total at 40 billion tonnes, with America in the lead, producing 3 million tonnes domestically every year, followed by China with 2.3 million tonnes."

Electronic waste contributes about 70 per cent of hazardous waste globally. Computer waste is an important component of e-waste and it is an alarming issue for global environment. In the US, 24 million computers become obsolete on an average every year. A report predicts that by 2020,